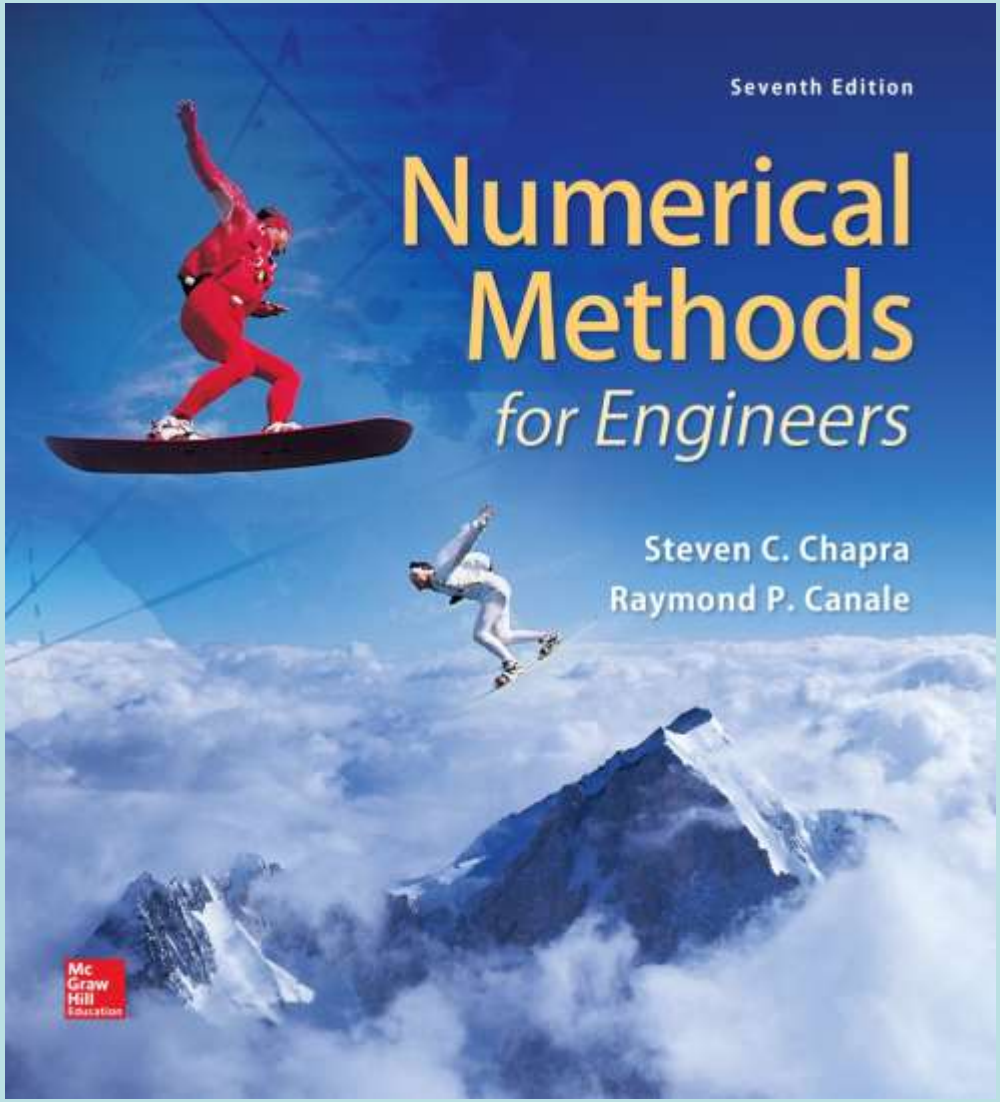
Part 4 - Chapter 13



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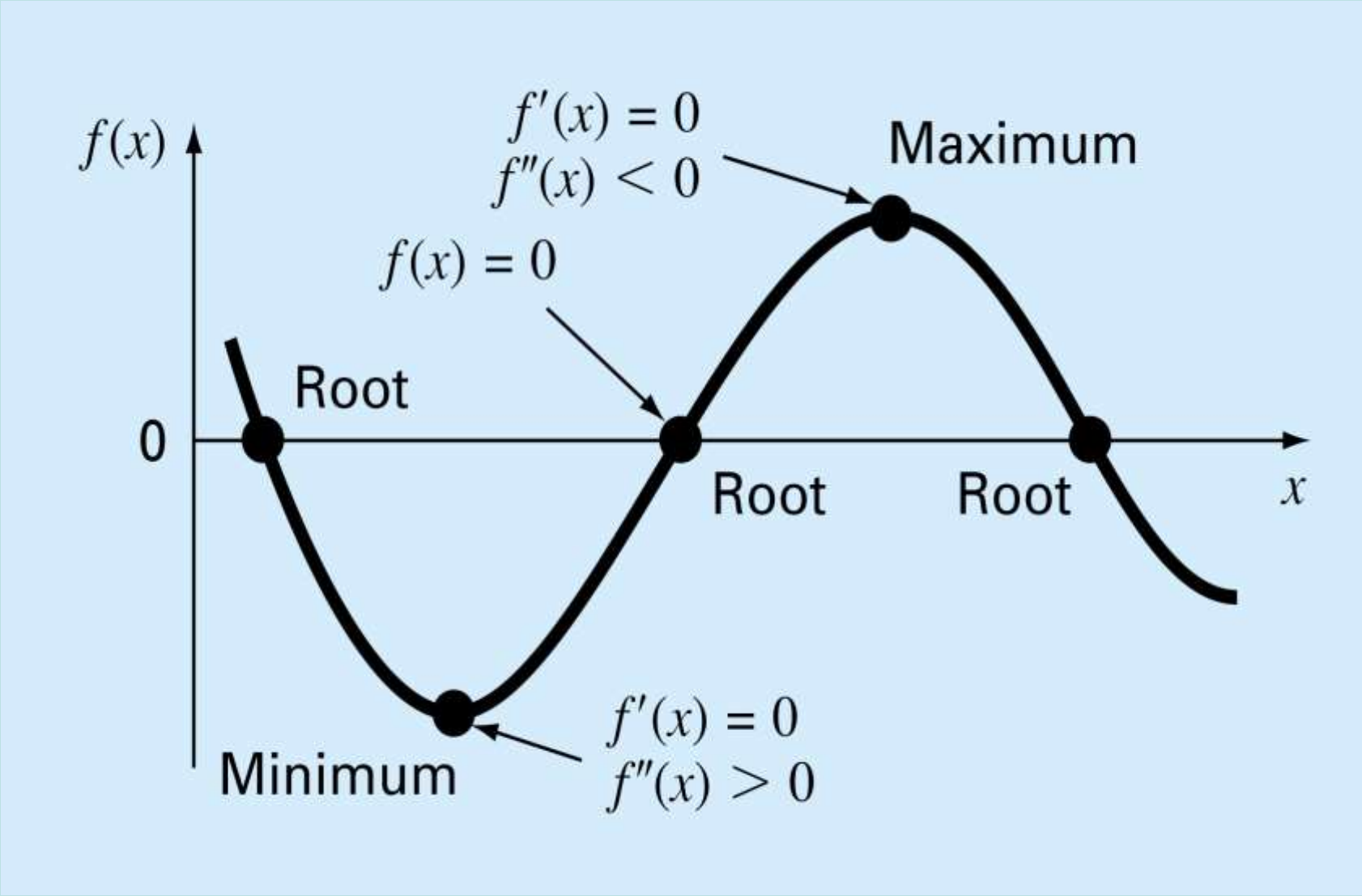
Optimization

Part 4

- Root finding and optimization are related, both involve guessing and searching for a point on a function.
- Fundamental difference is:
  - Root finding is searching for zeros of a function or functions
  - Optimization is finding the minimum or the maximum of a function of several variables.

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figure PT4.1



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Mathematical Background

- An *optimization* or *mathematical programming* problem generally be stated as:

  Find x, which minimizes or maximizes $f(x)$ subject to

$$d_i(x) \le a_i \quad i = 1,2,\ldots,m*$$
$$e_i(x) = b_i \quad i = 1,2,\ldots,p*$$

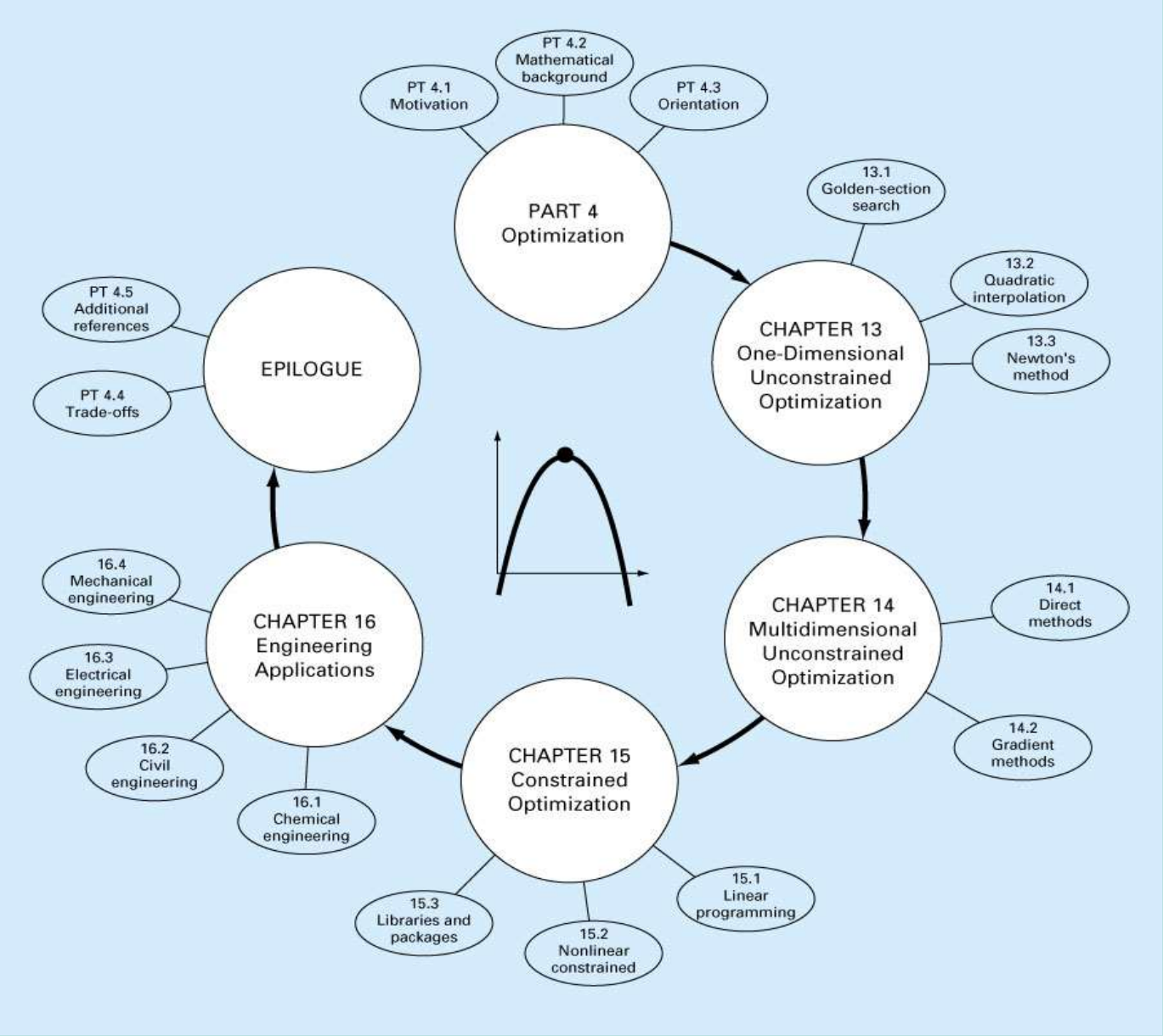
Where x is an n-dimensional *design vector*, $f(x)$ is the *objective function*, $d_i(x)$ are *inequality constraints*, $e_i(x)$ are *equality constraints*, and a_i and b_i are constants

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- Optimization problems can be classified on the basis of the form of *f(x):*
  - If *f(x)* and the constraints are linear, we have linear programming.
  - If *f(x)* is quadratic and the constraints are linear, we have quadratic programming.
  - If *f(x)* is not linear or quadratic and/or the constraints are nonlinear, we have nonlinear programming.
- When equations(*) are included, we have a *constrained optimization* problem; otherwise, it is *unconstrained optimization* problem.

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Figure PT4.5



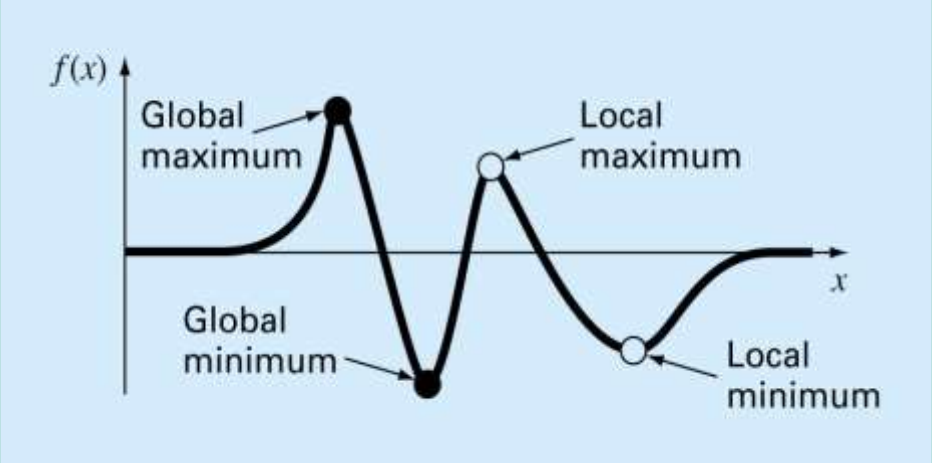
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One-Dimensional Unconstrained Optimization

Chapter 13

- In *multimodal* functions, both local and global optima can occur. In almost all cases, we are interested in finding the absolute highest or lowest value of a function.

Figure 13.1



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How do we distinguish global optimum from local one?

- By graphing to gain insight into the behavior of the function.
- Using randomly generated starting guesses and picking the largest of the optima as global.
- Perturbing the starting point to see if the routine returns a better point or the same local minimum.

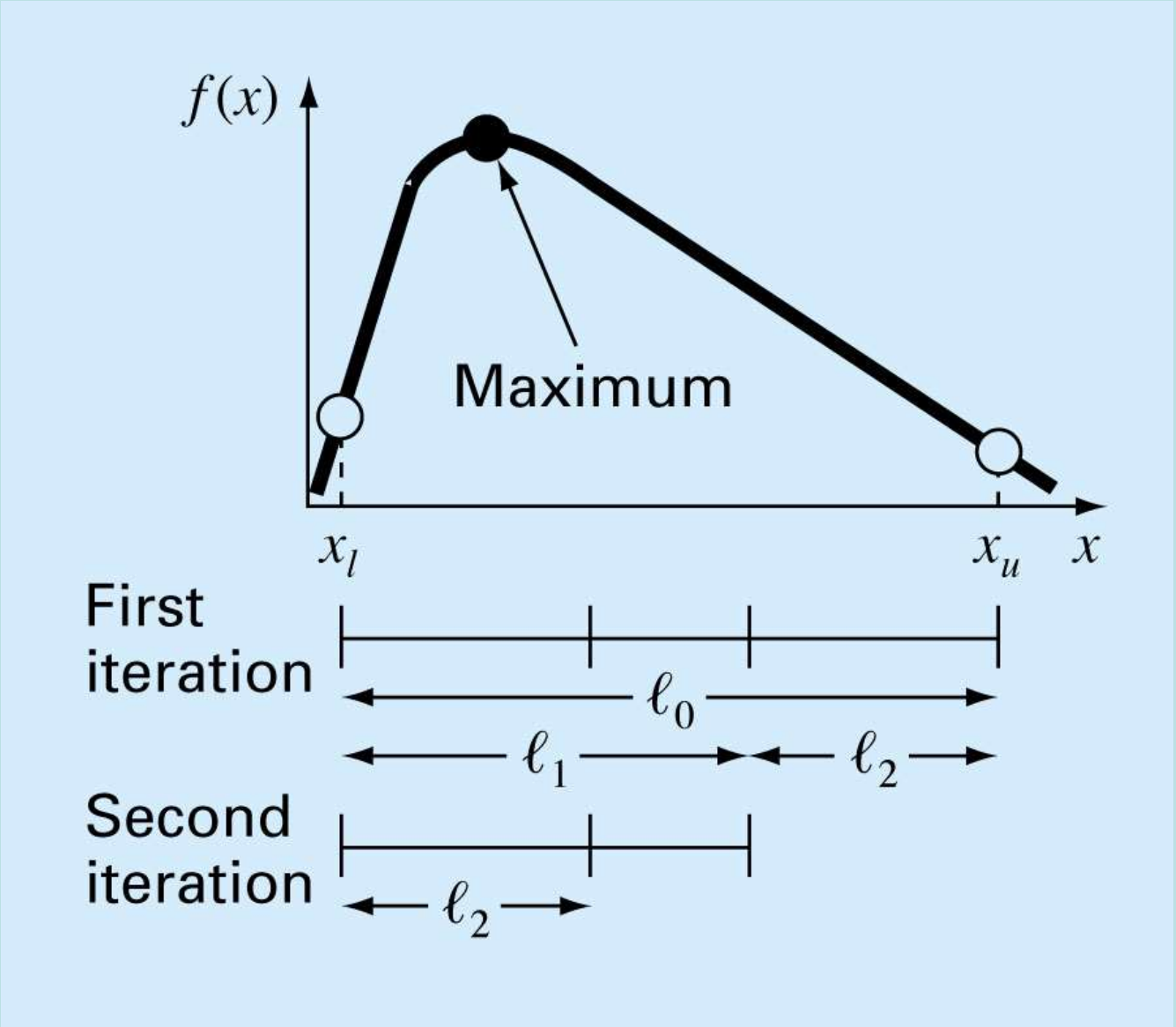
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Golden-Section Search

- A *unimodal* function has a single maximum or a minimum in the a given interval. For a *unimodal* function:
  - First pick two points that will bracket your extremum [x_l, x_u].
  - Pick an additional third point within this interval to determine whether a maximum occurred.
  - Then pick a fourth point to determine whether the maximum has occurred within the first three or last three points
  - The key is making this approach efficient by choosing intermediate points wisely thus minimizing the function evaluations by replacing the old values with new values.

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Figure 13.2



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$$l_0 = l_1 + l_2$$

$$\frac{l_1}{l_0} = \frac{l_2}{l_1}$$

- The first condition specifies that the sum of the two sub lengths l_1 and l_2 must equal the original interval length.

- The second say that the ratio of the length must be equal

$$\frac{l_1}{l_1 + l_2} = \frac{l_2}{l_1} \qquad R = \frac{l_2}{l_1}$$

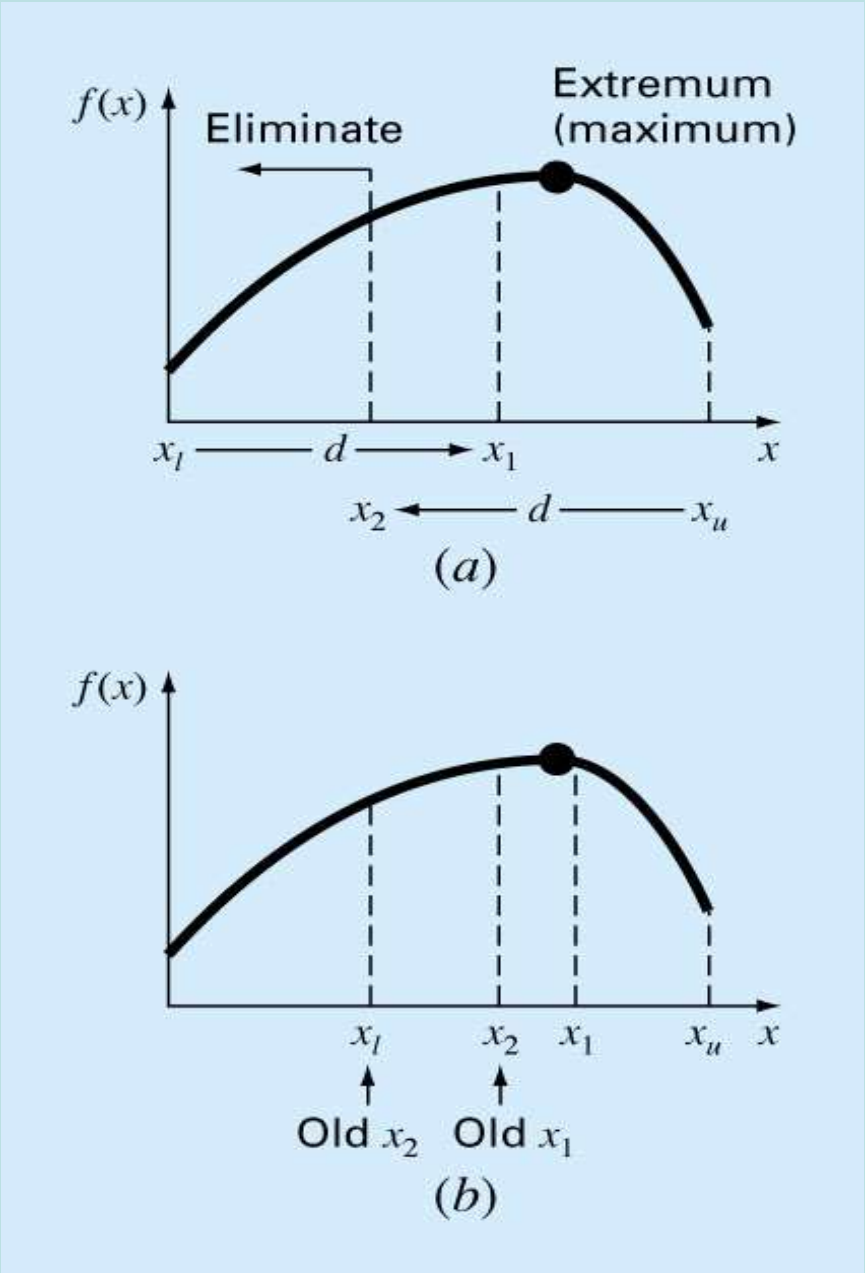
$$1 + R = \frac{1}{R} \qquad R^2 + R - 1 = 0$$

$$R = \frac{-1 + \sqrt{1 - 4(-1)}}{2} = \frac{\sqrt{5} - 1}{2} = 0.61803$$

Golden Ratio

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Figure 13.4



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The method starts with two initial guesses, x_l and x_u, that bracket one local extremum of *f(x)*:

- Next two interior points x_1 and x_2 are chosen according to the golden ratio

$$d = \frac{\sqrt{5}-1}{2}(x_u - x_l)$$

$$x_1 = x_l + d$$

$$x_2 = x_u - d$$

- The function is evaluated at these two interior points.

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Two results can occur:

- If $f(x_1)>f(x_2)$ then the domain of x to the left of x_2 from x_l to x_2, can be eliminated because it does not contain the maximum. Then, x_2 becomes the new x_l for the next round.

- If $f(x_2)>f(x_1)$, then the domain of x to the right of x_1 from x_l to x_2, would have been eliminated. In this case, x_1 becomes the new x_u for the next round.

- New $x_{1\ i}$s determined as before

$$x_1 = x_l + \frac{\sqrt{5}-1}{2}(x_u - x_l)$$

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- The real benefit from the use of golden ratio is because the original x_1 and x_2 were chosen using golden ratio, we do not need to recalculate all the function values for the next iteration.

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Newton's Method

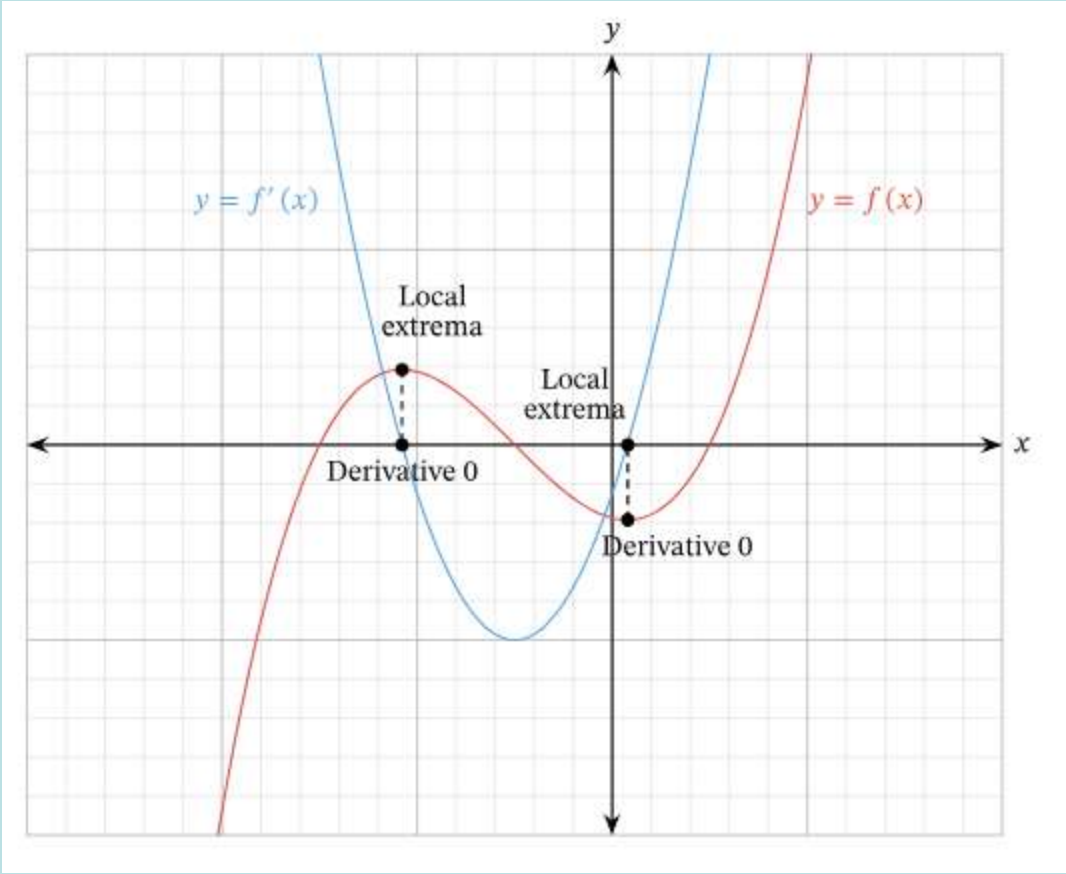
- A similar approach to Newton- Raphson method can be used to find an optimum of f(x) by defining a new function g(x)=f'(x). Thus because the same optimal value x* satisfies both

  $f'(x^*)=g(x^*)=0$

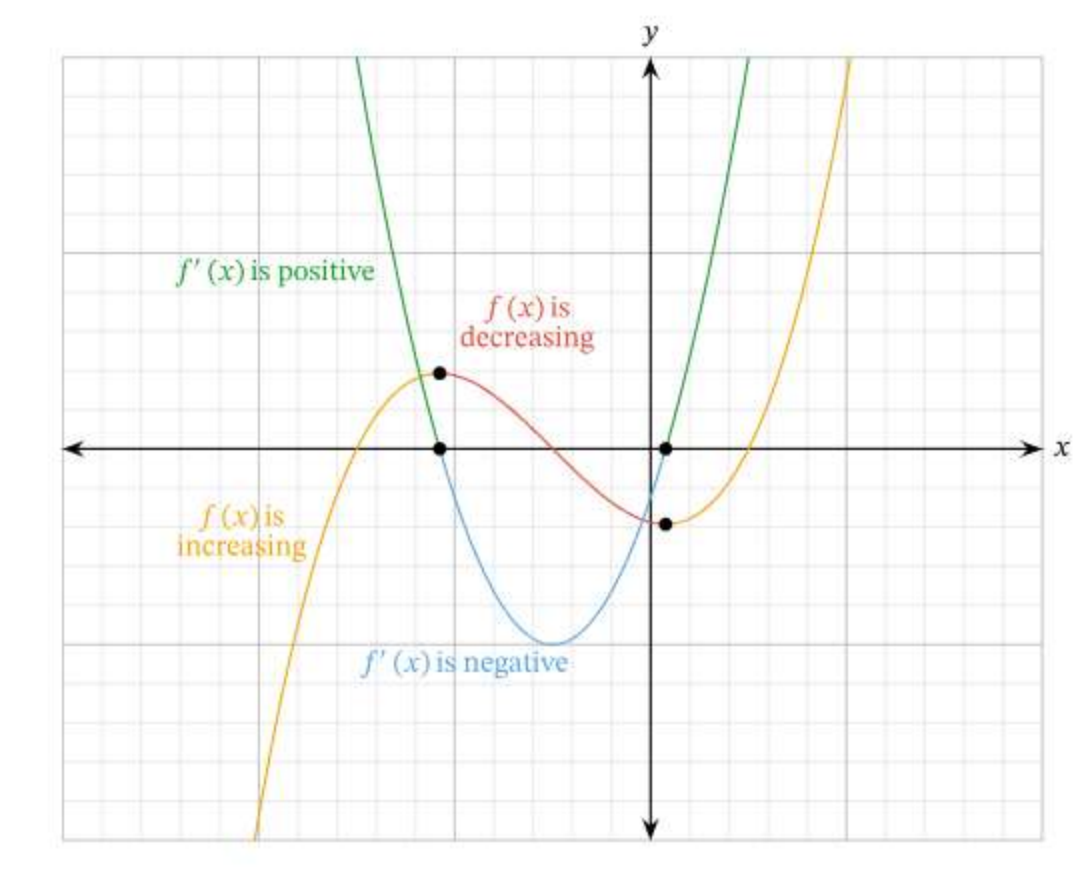
  We can use the following as a technique to the extremum of *f(x)*.

$$x_{i+1} = x_i - \frac{f'(x_i)}{f''(x_i)}$$

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